

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second Semester of M.Sc.(Mathematics) Examination May 2018
MA 724:Partial Differential Equations

Date: 10/05/2018,Thursday **Time:** 10:00 a.m. to 10:45 a.m. **Maximum Marks:** 20

Multiple Choice Questions

Instructions:

- (1) Tick the correct answer only in this sheet.
 - (2) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (3) Each question carries one mark.
 - (4) In this question paper, $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{Z}$ denotes the set of all integers. All notations are standard.
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[20]

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8. The general solution of the partial differential equation $(D^3 - 4D^2D' + 4DD'^2)z = 0$ is _____, where f, g and h are arbitrary functions.
- $z = f(y) + g(y - 2x) + xh(y - 2x)$
 - $z = f(y) + g(y + 2x) + xh(y + 2x)$
 - $z = f(x) + g(y - 2x) + h(y - 2x)$
 - $z = f(x) + g(y + 2x) + h(y - 2x)$.
9. The particular integral of the partial differential equation $(2D - D')(D^2 - D')z = e^{3x+y}$ is _____.
- ye^{3x+y}
 - $\frac{e^{3x+y}}{y}$
 - $\frac{e^{3x+y}}{40}$
 - $40e^{3x+y}$.
10. $\frac{D}{D'}(x^2 + y) =$ _____.
- x
 - y
 - $2x$
 - $2xy$
11. The partial differential equation $\frac{\partial^2 u}{\partial x^2} = 9 \frac{\partial^2 u}{\partial x \partial y}$ is _____.
- elliptic in the xy-plane
 - parabolic in the xy-plane
 - hyperbolic in the xy-plane
 - elliptic for $x > 0, y > 0$.
12. Which of the following statements is not true for the partial differential equation $y \frac{\partial^2 u}{\partial x^2} + (x + y) \frac{\partial^2 u}{\partial x \partial y} + x \frac{\partial^2 u}{\partial y^2} = 0$?
- The partial differential equation is parabolic for $x = y$
 - The partial differential equation is parabolic for $x \neq y$
 - The partial differential equation is not elliptic
 - The partial differential equation is hyperbolic for $x < y$.
13. In Riemann method of solution of general linear hyperbolic equation of the second order $L(z) = f(x, y)$, the adjoint operator to the operator $L(z) = \frac{\partial^2 z}{\partial x \partial y} + x \frac{\partial z}{\partial x} + x^2 e^y z$ is _____, where w be another function with continuous derivatives of the first order.
- $Mw = \frac{\partial^2 w}{\partial x \partial y} + x \frac{\partial w}{\partial x} + x^2 e^y w$
 - $Mw = \frac{\partial^2 w}{\partial x \partial y} + x \frac{\partial w}{\partial x} - x^2 e^y w$
 - $Mw = \frac{\partial^2 w}{\partial x \partial y} - \frac{\partial(xw)}{\partial x} - x^2 e^y w$
 - $Mw = \frac{\partial^2 w}{\partial x \partial y} - \frac{\partial(xw)}{\partial x} + x^2 e^y w$.
14. The characteristic of the differential equation $r - 4s + 4t = 0$ is _____.
- $x + 2y$
 - $2x + y$
 - $x^2 + y$
 - $x + y^2$.

15. _____ is the method of finding one or two intermediate integrals from a second order partial differential equation $f(x, y, z, p, q, r, s, t) = 0$.

- | | |
|---------------------|-------------------|
| a. Riemann's method | c. Monge's method |
| b. Voltera's method | d. none of these. |

16. If both the ends of a bar of some length are insulated and the initial temperature is given, then the temperature u in the bar is governed by _____, where k is a constant.

- | | |
|--|--|
| a. $\frac{\partial^2 u}{\partial t^2} = k \frac{\partial^2 u}{\partial x^2}$ | c. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ |
| b. $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ | d. none of these |

17. If the second order partial differential equation is hyperbolic, then we get _____.

- two distinct real families of characteristics
- only one family of characteristic
- two complex families of characteristics
- no family of characteristic.

18. The steady state temperature u in a rectangular plate in xy -plane is governed by _____.

- | | |
|--|--|
| a. $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ | c. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ |
| b. $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ | d. none of these |

19. Consider the boundary value problem:

$$\text{Laplace equation: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, 0 \leq x \leq a, 0 \leq y \leq b$$

$$\text{Boundary conditions: } u(0, y) = 0, u(a, y) = 0, 0 \leq y \leq b,$$

$$u(x, b) = 0, 0 \leq x \leq a \text{ and } u(x, 0) = f(x).$$

This problem is called_____.

- | | |
|------------------------|------------------|
| a. Dirichlet's problem | c. mixed problem |
| b. Neumann's problem | d. none of these |

20. Which of the following is two dimensional wave equation?

- | | |
|--|--|
| a. $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ | c. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ |
| b. $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ | d. $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ |

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MA 724 Partial Differential Equations

Date: 10/05/2018, Thursday **Time:** 10:45 a.m. to 01:00 p.m. **Maximum Marks:** 50

Instructions:

- (5) Section I and II must be written in separate answer sheets
 - (6) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (7) Figures to the right indicate marks.
 - (8) In this question paper, all notations are standard.
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SECTION-I

Q-2(a) Attempt any TWO of the following: [08]

- (i) Find the complete integral of $x^2p^2 + y^2q^2 = 4$.
- (ii) Solve the partial differential equation $(D^2 - 2DD' + D'^2)z = \cos(2x + y)$.
- (iii) Prove that, for the equation $\frac{\partial^2 z}{\partial x \partial y} + \frac{z}{4} = 0$, the Green function is $w(x, y, \xi, \eta) = J_0(\sqrt{(x - \xi)(y - \eta)})$, where $J_0(z) = \sum_{r=0}^{\infty} \frac{(-1)^r}{(r!)^2} \left(\frac{z}{2}\right)^{2r}$ is Bessel's function of first kind of order zero.

(b) Attempt any FOUR of the following: [12]

- (i) Solve the partial differential equation $z^2 + zu_z - u_x^2 - u_y^2 = 0$.
- (ii) Solve the partial differential equation $\left(\frac{y^2z}{x}\right)\frac{\partial z}{\partial x} + xz\frac{\partial z}{\partial y} = y^2$.
- (iii) Prove that $F(D, D')(e^{ax+by}) = F(a, b)e^{ax+by}$, where a and b are constants.
- (iv) Classify the partial differential equation $y^2r - x^2t = 0$ and also find its characteristics.
- (v) Using method of separation of variables, solve the boundary value problem $\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial y}$, if $u(0, y) = 8e^{-3y}$.

- (vi) What is boundary value problem? State Neumann boundary value problem in rectangle.

SECTION II

Q-3(a) Attempt any **THREE** of the following [18]

- (i) Show that the necessary and sufficient condition for the integrability of $dz = \phi(x, y, z) dx + \psi(x, y, z) dy$, where $p = \phi(x, y, z)$ and $q = \psi(x, y, z)$ are obtained by solving partial differential equations $f(x, y, z, p, q) = 0$ and $g(x, y, z, p, q) = 0$, is that $[f, g] = 0$.
- (ii) Solve the partial differential equations $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$.
- (iii) Show that a necessary and sufficient condition that the Pfaffian differential equation $\vec{X} \cdot \vec{dr} = P(x, y, z) dx + Q(x, y, z) dy + R(x, y, z) dz = 0$ be integrable is that $\vec{X} \cdot \text{curl} \vec{X} = 0$.
- (iv) Using Monge's method solve the partial differential equation $3s + rt - s^2 = 2$.
- (v) Reduce $x^2r + 2xys + y^2t = 0$ to canonical form and hence solve it.

(b) Attempt any **TWO** of the following [12]

- (i) Solve one dimensional wave equation $\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 y}{\partial t^2}$, where c is a constant satisfying the boundary conditions $y(0, t) = 0$ and $y(a, t) = 0$.
- (ii) If both ends of a bar of length a are at temperature zero and the initial temperature of a bar is $f(x)$, then find the temperature at any time t .
- (iii) Find the steady state temperature distribution in a rectangular plate of sides a and b insulated at the lateral surface and satisfying the boundary conditions $u(0, y) = u(a, y) = 0$, for $0 \leq y \leq b$, $u(x, b) = 0$ and $u(x, 0) = f(x)$ for $0 \leq x \leq a$.

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